# WATER OPTIMIZATION REPORT

SPB Erode Paper Mill

Q1 2025-26 Analysis

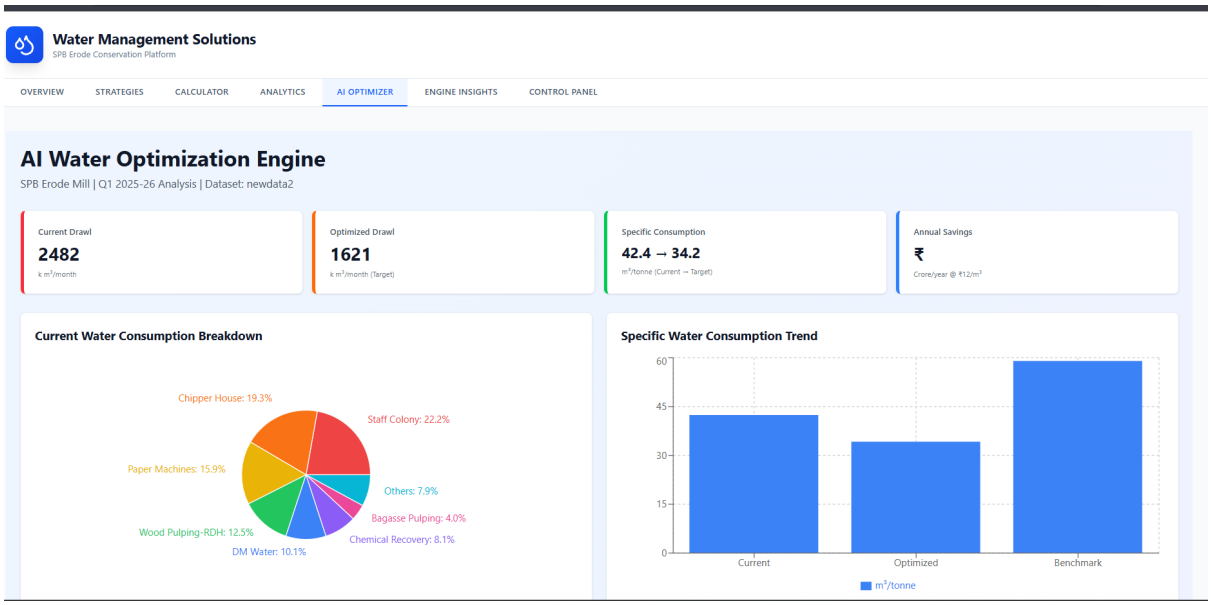
April 2026

## Executive Summary

**Current Situation:** SPB Erode Paper Mill consumes **2,482,000 $m^3$/month** with specific consumption of **42.4 $m^3$/tonne**. This results in an estimated annual water cost of **357–447 Crore** with **81.9% untapped optimization potential**.

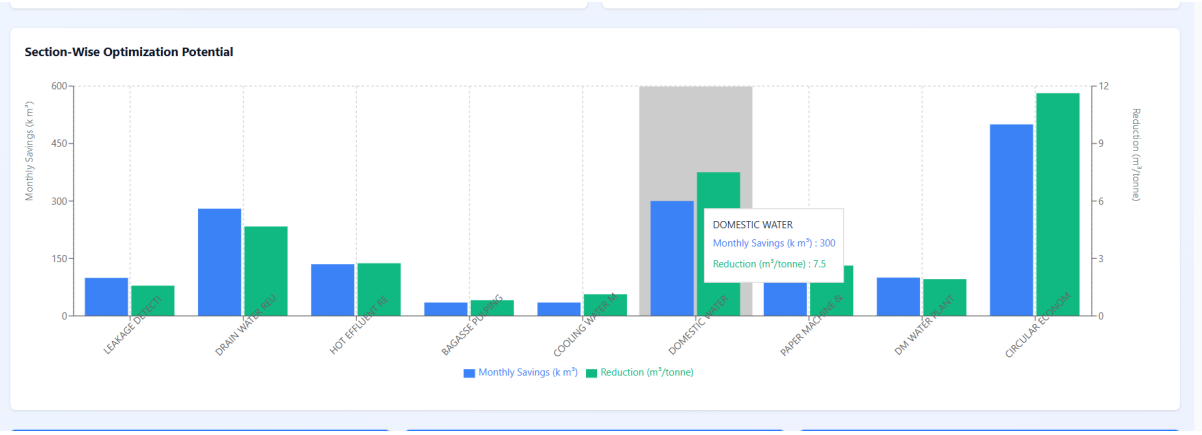
**Optimization Solution:** The AI Water Optimization Engine identifies **9 actionable strategies** to reduce consumption to **34.2 $m^3$/tonne** (42% better than industry benchmark), achieving **160–200 Crore annual savings** with a **2.5–3.5 year payback period**.

## Dashboard 1: AI Water Optimization Engine



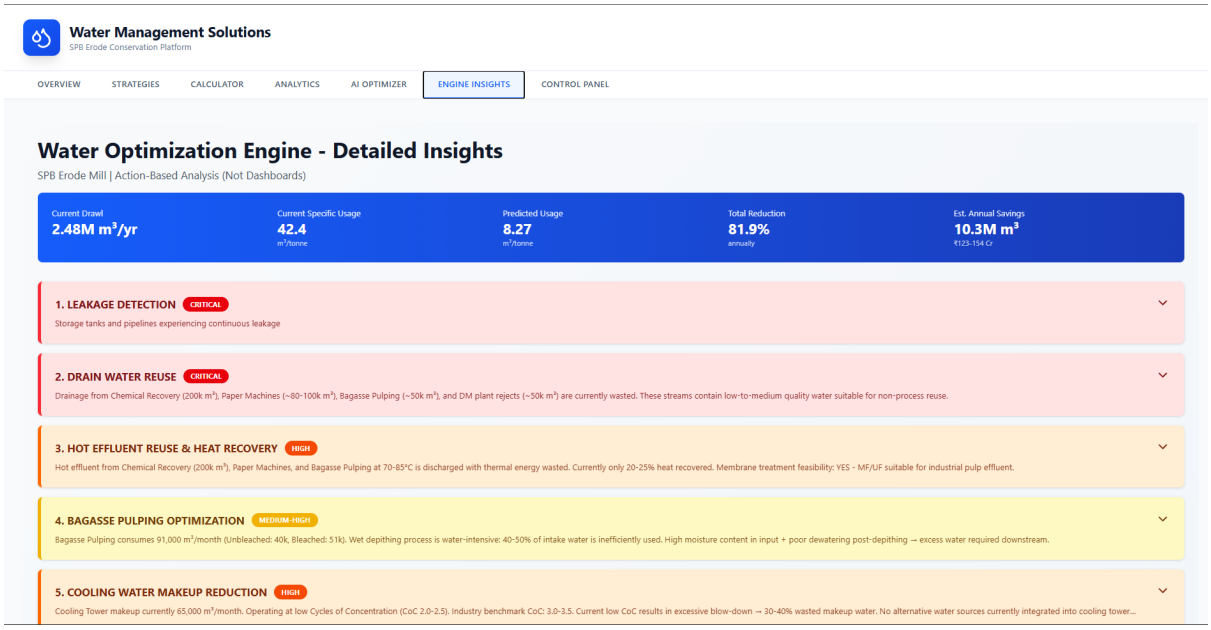

*Shows current water drawal (2,482k $m^3$/month), optimization target (1,621k $m^3$), reduction in specific consumption (42.4 $\rightarrow$ 34.2 $m^3$/tonne), and annual savings potential (160–200 Cr).*

## Dashboard 2: Section-wise Optimization Potential



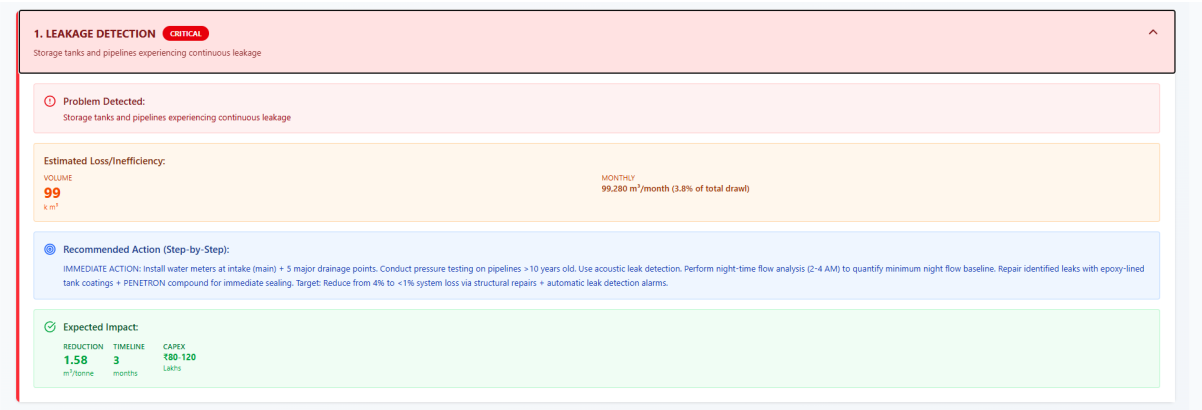

*Nine optimization areas ranked by monthly savings and consumption reduction. Domestic Water and Circular Economy show the highest impact.*

# Dashboard 3: Detailed Optimization Engine Insights




*Displays detailed insights across nine optimization areas including Leakage Detection, Drain Water Reuse, Hot Effluent Recovery, Bagasse Pulping, Cooling Water, Domestic Usage, Paper Machines, DM Water, and Circular Economy.*

# Dashboard 4: Leakage Detection - Critical Priority




*This dashboard represents one example from an interactive system where all optimization problems are structured using dropdown menus.*

**Explanation:**
The system is designed with a **dropdown-based interface** to manage multiple optimization areas efficiently.

- Each problem area is listed as a dropdown item
- Users can expand any item to view:
  - Problem identification
  - Loss estimation
  - Recommended actions
  - Expected impact
- The current image shows **Leakage Detection** as one example case

**Purpose:** This approach improves usability by making complex industrial data structured, interactive, and easy to navigate.